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DEBATE



The Second Drone Age: defining war in the 2020s

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Transformations, revolutions, evolutions, call them what you will. The simple fact is that the global proliferation of weaponised drones – known as the Second Drone Age – has altered the face of warfare in the 2020s. This is not to say that theoretical debates about what constitutes a true “revolution” in war are unimportant; more that the purpose of this article is to outline the clear ways in which armed drones have altered the practice of contemporary warfare. The broader academic debates will be left to those who wish to wage what Colin Gray called “protracted internecine combat” across the old “happy hunting ground” of “revolutionary” debates.¹ Instead, by focusing on the context specific empirics of armed drone deployments, by both state and non-state actors, key characteristics of the Second Drone Age will be outlined, and the growing impact of drones in warfare documented.

When researching drones, it is easy to get stuck in narrow discussions about specific types of Uncrewed Aerial Systems (UAS). The most common type of drone focused upon within academic literature is the Medium-Altitude Long-Endurance (MALE) drone and specifically the Western pioneering of such systems between the late-1980s and mid-2000s.² Such a focus is understandable. Armed and unarmed MALE drones – like the Predator class – have risen to become the spearhead of Western force deployment in the post-Cold War world.³ The trouble is, by cognitively restricting oneself to focusing on one form of drone technology and one military culture, a multitude of novel drone technologies and their strategic impact in conflicts around the world will be neglected. It is only by broadening the analytical scope to include non-Western cases of drone adoption (and deployment) that the true impact of drones on the practice of war can be understood.

Take the Tigray War in Ethiopia (2020–2022) as a pertinent example of how drones are altering warfare in the 2020s. When reflecting on the impact of Turkish TB2 and Chinese Wing Loong II drones in the Second Libyan Civil War (2014–2020), Agnes Callamard and I argued that “high-tech armed drones help decide the fate of nations”.⁴ Since then, that statement has generated robust debate about just how influential drones can be when it comes to ensuring state survival.⁵ Nevertheless, in 2021 the role drones can fulfil in securing a national government under threat from a powerful non-state group, became clear for all to see. After 12-months of prolonged conflict, the Ethiopian Prime Minister, Abiy Ahmed, altered his regime’s fortune by deploying armed drones from the UAE (Wing Loong IIs), Turkey (TB2s), and Iran (Mohajer-6s) against Tigray Rebels.⁶ In what the New York Times called “a stunning victory”, Ethiopian forces were able to

harness the drone's firepower to overpower an armed convoy of Tigray rebels heading to the capital to overthrow the government.⁷ As the Tigray leader, Debretsion Gebremichael, stated, it was "the drones provided by foreign powers" that caused him to order an immediate withdrawal from the attack.⁸ Drones were not solely responsible for this change in fortune. The competence of the Ethiopian forces on the ground and the nation's ongoing efforts to cut supply lines to the rebels contributed to the routing of the Tigray forces. Nevertheless, the simple fact is that armed drones proved decisive. If they had not been available to the Abiy government, then today there would be a very different regime running Ethiopia.

However, it is not just the proliferation of drones to sure up struggling states that defines the Second Drone Age.⁹ Although current figures show 113 nation states have a military drone programme (an increase from 60 in 2010), there are now an estimated 65 non-state groups that have weaponised drones.¹⁰ One of the most prominent groups are the Houthi Rebels in Yemen. As a terrorist organisation supported by elements of the Iranian state, the Houthi Rebels were able to procure and harness increasing levels of high-tech Iranian drone technologies.¹¹ Systems like the Qassef-2K loitering munition and the Samad-3 long-range armed UAV (1500km), reportedly supplied to the Houthis, allow this non-state group to attack Iran's enemies (and their own enemies) with military grade precision-weapons technologies.¹² These drones are similar to those supplied by Iran to Russia in 2022 and 2023 to use against Ukraine and have been utilised in a similar way.¹³ Just as President Putin ordered long-range drone attacks against Ukraine's energy infrastructure and its major civilian centres, so too have the Houthis across the Middle-East. The infamous attacks on the ARAMCO oil processing sites in Saudi Arabia (2019) and Abu Dhabi International Airport in the UAE (2021), highlight the fact that non-state groups are now able to deploy airpower in a manner that was once reserved for powerful state militaries.¹⁴ A novel addition to conflict.

Still, this is not the only way the Houthi use of drones in altering warfare. In 2019, I inspected captured Houthi drones in the Middle East and found that the Houthi drones were not solely state supplied systems. Instead, best described as Hybrid Drones, the Houthi's have used the limited number of state-supplied Iranian systems as moulds to recast additional numbers of identical fibreglass shells. These shells, as aerodynamic as the state design, are then augmented with commercially available elements – such as engines from Europe, wiring from China, and various commercial drone components – turning the shells into deployable weapons system. The interesting thing to note here is that all these recast drones are produced locally, meaning that the Houthis can operate lethal airpower without the supply and direction of the Iranian state. If supplies are cut off, or political ties fray, they can simply supply themselves (and any allied terrorist organisation they so wish) with armed drones. With so many state and commercial organisations now producing drones, the Houthis have a wide, adaptable, and hard to stop supply line. As such, through the adoption of both state and commercial drones, non-state actors have found a way to evade technology control regimes, become resilient against embargos, and create a robust supply chain that will facilitate their mastering of the air.¹⁵

Overall, drones are defining (and redefining) the practice of warfare globally in the 2020s. All future wars will be drone wars in the fact that one side of the conflict – or

more likely both – will have an arsenal of commercially sourced and/or state manufactured drone systems. While work is underway at the United Nations to build upon the 2022 Delhi Declaration that seeks to stem “the increasing global misuse of unmanned aerial systems (UAS) ...”, it is unlikely that the current generation of drone technologies (and know-how) will be restricted.¹⁶ Instead, the cat is out of the bag. The Second Drone Age is with us. Drones are here to stay.

Notes

1. Colin S. Gray, *Recognizing and understanding revolutionary change in warfare: The sovereignty of context*. (Strategic Studies Institute, US Army War College, 2006).
2. Richard Whittle, *Predator: The Secret Origins of the Drone Revolution* (Picador, New York, 2015).
3. Paul Lushenko, ‘U.S. Presidents’ Use of Drone Warfare’, *Defense & Security Analysis* 38, no. 1 (2022): 31–52; James Rogers and Delina Goxho, ‘Light Footprint – Heavy Destabilising Impact in Niger: Why the Western Understanding of Remote Warfare Needs to be Reconsidered’, *International Politics* (published online January, 2022), <http://doi.org/10.1057/s41311-021-00362-9>.
4. James Rogers and Agnes Callamard, ‘We Need a New International Accord to Control Drone Proliferation’, *Bulletin of the Atomic Scientists*, 1 December 2020, <https://thebulletin.org/2020/12/we-need-a-new-international-accord-to-control-drone-proliferation/>.
5. Antonio Calcara, Andrea Gilli, Mauro Gilli, Raffaele Marchetti, and Ivan Zaccagnini, ‘Why Drones Have Not Revolutionized War: The Enduring Hider-Finder Competition in Air Warfare’, *International Security* 46, no. 4 (2022): 130–171.
6. Wim Zwijnenburg, ‘Turkish Drones Join Ethiopia’s War, Satellite Imagery Confirms’, *PAX*, 11 January 2022, <https://paxforpeace.nl/news/blogs/turkish-drones-join-ethiopias-war-satellite-imagery-confirms>.
7. Declan Walsh, ‘Foreign Drones Tip the Balance in Ethiopia’s Civil War’, *New York Times*, 20 December 2021, <https://www.nytimes.com/2021/12/20/world/africa/drones-ethiopia-war-turkey-emirates.html>
8. The Intra-State use of drones will be a point of growing concern throughout the 2020s. As states seek to sure up their borders, quash rebellions, and suppress public unrest, it is the drone that will allow the metropole to project power over thousands of kilometres to traditionally hard to reach regions. This will become an especially salient issue in regions under threat from extreme Climate Change. *Ibid*.
9. Ash Rossiter, ‘Drone Usage by Militant Groups: Exploring Variation in Adoption’, *Defense and Security Analysis* 34, no. 2 (2018): 113–26.
10. As I stated in my address to the UN Security Council on August 31st, 2022, this is an 88.2 percent increase in just 12 years. For the breakdown of the stats see Kerry Chávez and Ori Swed, ‘The Proliferation of Drones to Violent Nonstate Actors’, *Defence Studies* 21, no. 1 (2020): 1–24. Also see James Rogers, ‘Threats to International Peace and Security caused by Transnational Activities of Terrorist Groups’, Address to UN Security Council Arria-Formula (United Nations, 2022), <https://media.un.org/en/asset/k1q/k1qcqls4b9>.
11. Francesco F. Milan and Aniseh Bassiri Tabrizi, ‘Armed, Unmanned, and in High Demand: The Drivers Behind Combat Drones Proliferation in the Middle East’, *Small Wars & Insurgencies* 31, no. 4 (2020): 730–50.
12. James Rogers, ‘Future Threats: Military UAS, Terrorist Drones, and the Dangers of the Second Drone Age,’ in *A Comprehensive Approach to Countering Unmanned Aircraft Systems*, ed. NATO JAPCC (Kalkar, Germany: Joint Airpower Competence Center, 2020), 481–505.

13. Mohammad Eslami, 'Iran's Drone Supply to Russia and Changing Dynamics of the Ukraine War', *Journal for Peace and Nuclear Disarmament* 5, no. 2 (2022): 507–18.
14. It is important to note that weaponised drones and their precision strike capabilities have also been used to target international shipping, US and allied military bases, and for the assassination of key military and political figure. All of which will likely increase in occurrence as drones spread further (and in greater numbers) through the 2020s. James Rogers, 'Last Month, Three Drones Attacked an Israeli Tanker: Here's Why That's Something New', *Washington Post*, 19 August 2021, <https://www.washingtonpost.com/politics/2021/08/19/last-month-three-drones-attacked-anisraeli-tanker-heres-why-thats-something-new/>.
15. By deploying terror from thousands of kilometres away, terrorist organisations are able to evade the risks of crossing national borders and transcend the international counter-terrorist measures that have been set up post-9/11 to detect and detain them. Instead, terrorists can now conduct attacks across borders, by remote control, and from the comfort of their distant and well-defended safe havens. In fact, if current generation of state drones – like the Chinese Wing Loong III with its 10,000km range – were to fall into non-state hands, they would introduce a novel proliferation of intercontinental threats that would make global reach and global strike accessible to all. See James Rogers, 'The Third Drone Age: Vision Out to 2040', *Center for International Governance Innovation*, 28 November 2022, <https://www.cigionline.org/articles/the-third-drone-age-visions-out-to-2040/> and James Rogers and Arthur Holland Michel, 'Drone Warfare: Distant Targets and Remote Killings', in *Palgrave Encyclopaedia of Global Security Studies* (2020) https://link.springer.com/referenceworkentry/10.1007/978-3-319-74336-3_33-1.
16. UNCTC, *Delhi Declaration on Countering the Use of New and Emerging Technologies for Terrorist Purposes* (UN CTC, 2022).

Disclosure statement

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